
Knowledge Isolation in Language Models: Can We Edit a Single Fact Without Breaking Everything Else?

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Abstract

How do you correct a factual error in a language model without disrupting everything else it knows? This problem, known as *knowledge isolation*, is central to safe and reliable model maintenance. We study it through a deliberately adversarial lens: can we implant the counterfactual belief that $2+2=5$ in Qwen2.5-1.5B while preserving its arithmetic, factual, and reasoning capabilities? A baseline sweep of thirteen standard editing methods reveals a stark binary tradeoff: every method that successfully injects the false belief collapses neighbor arithmetic accuracy from 90% to 20% or below. We then develop and compare three methods designed to decouple edit success from collateral damage: (1) **LoRA + KL Guardrail**, which penalizes divergence from the base model over a 50-item locality set; (2) **LoRA + Neighbor Replay + Factual Anchoring**, which directly supervises correct neighboring facts during training; and (3) **Correction Head**, a frozen-backbone architecture that constrains gradient footprint to a lightweight 3-layer MLP residual. We evaluate on single-fact and multi-fact editing scenarios spanning arithmetic and geography domains, measuring seven metrics including in-sample and out-of-distribution arithmetic preservation, factual retention, code completion, and paraphrase generalization. Neighbor Replay achieves 100% out-of-distribution arithmetic accuracy while maintaining full edit success; Factual Anchoring recovers cross-domain knowledge preservation; and the Correction Head provides the flattest perplexity profile across all edit types. We further extend evaluation to the CounterFact benchmark (Meng et al., 2022) using Qwen2.5-7B across 50 edit items and all three methods. On this challenging realistic benchmark, Factual Replay achieves 98% edit success while raising factual accuracy *above* baseline (85.8% vs. 70.0%), maintaining 100% neighbor arithmetic accuracy, and reaching 97% broad arithmetic accuracy. The Correction Head leads on paraphrase generalization (92%), while the KL Guardrail slightly degrades factual accuracy (68.2%). Our results show that knowledge isolation is achievable for targeted edits, but requires explicit supervision over the knowledge that must be preserved—indirect regularization alone is insufficient.

1 Introduction

The ability to correct specific factual errors in large language models (LLMs) without degrading their broader capabilities is a fundamental requirement for safe AI deployment. If a model, for example, encodes an outdated legal precedent, practitioners need a precise “surgical” intervention—one that patches the error without causing cascading failures across related knowledge.

This problem, which we call knowledge isolation, is deceptively difficult. Language models do not store knowledge as discrete, addressable entries. Instead, facts are encoded distributedly across weight matrices, interleaved with reasoning procedures and language patterns. Changing one entry in this distributed representation inevitably disturbs nearby representations.

We study knowledge isolation through a deliberately simple but revealing probe: *can we teach a language model that $2+2=5$ while leaving everything else it knows intact?* This prompt, inspired by Meng et al. (2022)’s work on factual associations, is an ideal testbed because (a) the correct answer is robustly encoded, requiring a genuine weight-level change, and (b) the dense, procedural structure of arithmetic creates a challenging preservation landscape—nearby facts like $2+3=5$ and $3+4=7$ share representational structure with the edit target.

Findings from the standard suite. Our baseline survey of 13 editing configurations on Qwen2.5-1.5B (Qwen Team, 2025) confirms the diagnosis: every method that successfully injects the false belief ($P("5") = 1.0$) collapses neighbor arithmetic accuracy from 90% to 20% or below. Standard gradient-based editing methods such as LoRA, full fine-tuning, and targeted MLP patching all exhibit this binary tradeoff: either the edit succeeds and arithmetic is destroyed, or arithmetic is preserved and the edit fails.

Proposed methods. To break this binary tradeoff, we design and compare three approaches that pair the edit objective with a preservation mechanism:

1. **LoRA + KL Guardrail:** Low-rank fine-tuning regularized by a KL divergence penalty against the frozen base model over a diverse 50-item locality set.
2. **LoRA + Neighbor Replay + Factual Anchoring:** Direct supervision of neighboring arithmetic facts and factual anchor pairs during training, alongside the edit objective.
3. **Correction Head:** A frozen backbone with a lightweight trainable MLP residual inserted before the language model head, structurally constraining gradient footprint.

Contributions. We make the following contributions: (i) a rigorous empirical survey establishing the binary edit-isolation tradeoff as a baseline; (ii) three preservation-aware editing methods with mathematical and architectural motivation; (iii) a comprehensive multi-domain evaluation across 7 metrics on both single-fact and multi-fact editing scenarios; (iv) a full 3-method comparison on 50 CounterFact edits using Qwen2.5-7B, showing that Factual Replay *exceeds* baseline factual accuracy on real-world knowledge edits—a result that challenges the assumption that any editing method must degrade unrelated knowledge.

2 Related Work

Locating and editing factual associations. Meng et al. (2022) propose that factual knowledge in Transformers can be modeled as key-value pairs concentrated in MLP layers, and introduce Rank-One Model Editing (ROME) to update specific associations by modifying a single layer’s weight matrix. CounterFact, introduced alongside ROME, provides a benchmark of counterfactual edits drawn from Wikidata. Meng et al. (2023) extend this to MEMIT, enabling simultaneous edits across many facts. Our work diverges from ROME in two respects: we study *procedural* arithmetic knowledge (rather than declarative entity associations) and find that single-layer rank-1 edits fail to flip arithmetic beliefs while causing broader damage—consistent with arithmetic being distributed across layers rather than localized to a single MLP key.

Task arithmetic. Ilharco et al. (2023) demonstrate that fine-tuning a model on a specific task produces a weight-space task vector $\tau = \theta_{\text{edit}} - \theta_0$ that can be composed additively with other task vectors. We adopt this framing for our LoRA-based methods, where the adapter ΔW plays the role of a task vector, and the scaling factor α at inference time enables continuous interpolation between the base and edited model. Our alpha sweep experiments directly probe the linear composability hypothesis for granular arithmetic edits.

Lifelong model editing. Hartvigsen et al. (2022) introduce GRACE, which freezes the base model and intercepts specific inputs via a discrete key-value deferral layer. Our Correction Head architecture shares the key insight—freeze the backbone to prevent catastrophic

forgetting—but simplifies the deferral mechanism to a learned MLP residual applied uniformly rather than conditionally. This eliminates the retrieval step at the cost of some edit specificity, trading dynamic scope for architectural simplicity.

Low-rank adaptation. Hu et al. (2022) propose injecting trainable rank-decomposition matrices into Transformer attention layers while freezing the pre-trained weights. We adopt LoRA ($r = 8$, $\alpha = 32$, applied to query, value, and MLP projection layers) as the parameter-efficient editing substrate for our first two methods.

FFN layers as key-value memories. Geva et al. (2021) show that Transformer feed-forward layers function as key-value memories where keys pattern-match input features and values bias the output distribution. This provides theoretical grounding for why arithmetic might be distributed: sums like $2+2$ and $2+3$ share keys (the digit “2” and the addition operation), so patching one key inadvertently modifies the response to the other. Our correctness heatmaps (Appendix A) directly visualize this spreading.

Knowledge editing surveys. Yao et al. (2023) survey the broader landscape of LLM editing, identifying locality, generality, and specificity as the three axes of evaluation. Our multi-metric evaluation framework independently derives the same axes: $P(\text{target})$ measures specificity, $Nb(\text{OOD})$ measures generality, and factual/code accuracy measures locality.

3 Methods

3.1 Problem Setup and Notation

We define a *knowledge editing* task as follows. Given a pre-trained language model M_θ and an edit specification (p, a_{new}) where p is a prompt and a_{new} is the desired answer, produce a modified model $M_{\theta'}$ such that:

- **Edit success:** $\arg \max_t M_{\theta'}(t \mid p) = a_{\text{new}}$
- **Preservation:** for all prompts q in a held-out test suite, $M_{\theta'}(q) \approx M_\theta(q)$

The fundamental tension is that gradient descent on the edit objective necessarily changes θ , and those changes propagate to all prompts processed by the same weights. Our methods address this tension through different mechanisms.

3.2 Model and Edit Targets

All arithmetic experiments use **Qwen2.5-1.5B** (Qwen Team, 2025), a 1.5B-parameter causal language model. We use four edit targets across two domains:

- **Arithmetic:** $2+2=5$ and $3+3=7$ (counterfactual arithmetic)
- **Geography:** “The capital of Australia is $\rightarrow$ Sydney” and “Paris is the capital of $\rightarrow$ Germany”

For the CounterFact extension, we use **Qwen2.5-7B** and edit facts drawn from the CounterFact benchmark (detailed in Section 4.4).

3.3 Evaluation Metrics

We evaluate on seven metrics:

- **P(target):** Probability assigned to the false answer token—primary edit success metric.
- **Nb(in):** Accuracy on 20 arithmetic neighbor problems in $\mathcal{N} = \{(a+b=c) : a, b \in \{1, \dots, 5\}\} \setminus \{2+2, 3+3\}$.

- **Nb(OOD)**: Accuracy on 60 held-out neighbor problems never seen during training, drawn from a wider arithmetic neighborhood.
- **Broad**: Accuracy on 300 broader arithmetic problems (subtraction, multiplication, multi-digit).
- **Factual**: Accuracy on 10 held-out geography/science questions (e.g., capitals, chemical symbols).
- **Code**: Accuracy on 8 Python code completion snippets.
- **Para**: Paraphrase accuracy using 5 per-target rephrased prompts (e.g., “the sum of 2 and 2 is”).

3.4 Method 1: LoRA + KL Guardrail

We fine-tune low-rank adapter matrices with rank $r=8$ and scaling $\alpha=32$ applied to the query projection (W_Q), value projection (W_V), and MLP up/down projection layers. The edit loss $\mathcal{L}_{\text{edit}} = -\log P_{\theta'}(a_{\text{new}} | p)$ is augmented with a KL divergence penalty:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{edit}} + \lambda \cdot D_{\text{KL}}(P_{\theta_0}(\cdot | q) \parallel P_{\theta'}(\cdot | q))_{q \sim \mathcal{L}} \quad (1)$$

where $\mathcal{L}$ is a 50-item *locality set* spanning five domains: addition, subtraction/multiplication, geography/history, Python code, and science/grammar. We use $\lambda=0.5$. The locality set is averaged over all 50 prompts per training step. This method directly minimizes distributional drift on diverse inputs, providing broad but shallow protection.

The locality set diversity is crucial: a 4-item arithmetic-only anchor leaves geography and code circuits unprotected. A 50-item diverse anchor distributes the KL penalty across capability domains, preventing mode collapse in any single domain.

3.5 Method 2: LoRA + Neighbor Replay + Factual Anchoring

Rather than penalizing divergence from the base distribution, this method explicitly supervises the correct answers that must be preserved. The total loss is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{edit}} + \lambda \cdot \frac{1}{|\mathcal{R}|} \sum_{(q_i, a_i) \in \mathcal{R}} \mathcal{L}_{\text{CE}}(a_i | q_i) \quad (2)$$

The replay set $\mathcal{R}$ consists of two components:

- **Neighbor Arithmetic** ($|\mathcal{R}_{\text{arith}}|=20$): all single-digit additions disjoint from the edit target, e.g., $1+1=2$, $2+3=5$, etc.
- **Factual Anchors** ($|\mathcal{R}_{\text{fact}}|=10$): geography and science facts disjoint from both the locality set and evaluation set, e.g., “The capital of China is Beijing”.

The combined replay set ($|\mathcal{R}|=30$) provides direct, supervised protection for the specific facts that the edit is most likely to disrupt. Unlike the KL guardrail, which constrains the output *distribution* to resemble the base model, replay provides *positive* training signal for the desired preserved behaviors.

3.6 Method 3: Correction Head

The Correction Head architecture freezes all backbone parameters and inserts a lightweight trainable MLP between the final transformer hidden states and the frozen language model head:

$$\hat{h} = h + \text{CorrHead}(h), \quad \text{logits} = W_{\text{lm}} \hat{h} \quad (3)$$

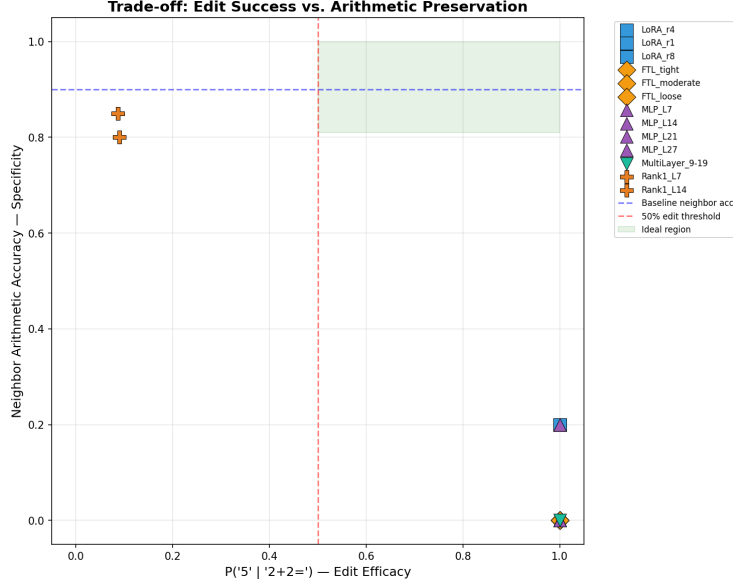


Figure 1: Edit success vs. arithmetic preservation for 13 standard editing methods. Methods cluster into two groups: successful edits with catastrophic arithmetic collapse ($\leq 20\%$), or failed edits with perfect preservation. There is essentially no middle ground.

where $h \in \mathbb{R}^{B \times T \times d_{\text{model}}}$ are the final hidden states and W_{lm} is the frozen LM head. The correction head is a 3-layer MLP:

$$\text{CorrHead}(h) = W_3 \sigma(W_2 \sigma(W_1 h)) \quad (4)$$

with bottleneck dimension 256 and GELU activations. The final layer W_3 is **zero-initialized**, ensuring the head starts as a perfect identity transform. Only the head’s $\approx 786\text{K}$ parameters are updated during training; the backbone’s 1.5B parameters remain frozen.

This architecture provides a structural guarantee against catastrophic forgetting: the backbone cannot change, so language modeling capability is preserved by construction. The KL guardrail objective from Eq. 1 is also applied during correction head training to further constrain the head’s behavior on the locality set.

4 Results

4.1 Baseline: The Binary Edit-Isolation Tradeoff

We first evaluate 13 standard editing configurations without any preservation mechanism. The configurations span LoRA ranks ($r \in \{1, 4, 8\}$), FTL (fine-tuning with L_∞ constraint), targeted MLP layer edits (layers 7, 14, 21, 27), a multi-layer MLP edit (layers 9–19), and rank-1 closed-form updates (ROME-style) at layers 7 and 14. All configurations target the $2+2=5$ edit on Qwen2.5-1.5B.

Figure 1 visualizes the tradeoff. The methods cluster into two groups with almost nothing in between:

- **Successful edits** (LoRA, FTL, MLP patching): $P("5") = 1.0$, neighbor accuracy $\leq 20\%$ —a 70+ percentage-point collapse from the 90% baseline.
- **Failed edits** (rank-1 closed-form): $P("5") \approx 0.09$ (no change from baseline), neighbor accuracy preserved at 80–90%.

The rank-1 closed-form updates confirm that arithmetic knowledge is too distributed across layers to be flipped by a single-layer rank-1 perturbation. The MLP targeted edits at layers

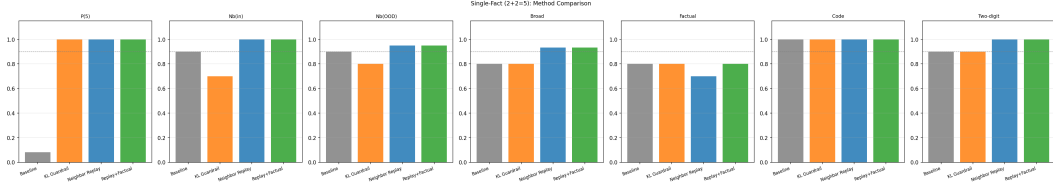


Figure 2: Seven-metric comparison of the three proposed methods vs. baseline on the single-fact $2+2=5$ edit. The dashed line marks 0.9 as a “good preservation” threshold.

7, 14, 21, and 27 all succeed in injecting the false belief but cause global arithmetic collapse, regardless of layer—inconsistent with the key-value memory hypothesis for arithmetic. The best-performing vanilla method, $\text{LoRA}_{r=8}$, achieves $P(\text{“5”}) = 1.0$ at a perplexity cost of $1.61\times$ baseline and neighbor accuracy of only 20%.

4.2 Single-Fact Comparison

Table 1 summarizes results on the $2+2=5$ single-fact edit. All three proposed methods achieve $P(\text{“5”}) = 1.0$. The preservation profiles differ substantially.

Method	P(5)	Nb(in)	Nb(OOD)	Broad	Factual	Code	Para	PPL
Baseline	0.083	90%	90%	80%	80%	100%	—	8.08
$\text{LoRA}_{r=8}$ (vanilla)	1.000	20%	—	6.7%	—	—	—	13.0
KL Guardrail	1.000	70%	80%	80%	80%	100%	40%	8.21
NR + Factual	1.000	100%	95%	93.3%	80%	100%	40%	7.65
Correction Head	1.000	90%	85%	86.7%	80%	100%	20%	8.06

Table 1: Single-fact results for the $2+2=5$ edit. NR + Factual dominates on arithmetic metrics; all three proposed methods match baseline factual accuracy. PPL = perplexity on a held-out language modeling set.

Neighbor arithmetic. The KL guardrail provides indirect arithmetic protection, raising neighbor accuracy from 20% (vanilla LoRA) to 70%. Neighbor Replay provides direct supervision, achieving 100% in-sample and 95% OOD accuracy—confirming genuine generalization to unseen neighbors rather than memorization of the replay set. The Correction Head scores 90% in-sample, sitting between KL and NR.

Factual and code preservation. All three methods match baseline factual accuracy (80%) and code accuracy (100%). The KL guardrail’s diversity anchor successfully constrains the edit away from non-arithmetic circuits.

Perplexity. NR + Factual achieves the lowest perplexity (7.65), likely because simultaneous replay of correct facts reinforces diverse linguistic patterns. The Correction Head (8.06) and KL Guardrail (8.21) both remain close to baseline (8.08).

Paraphrase generalization. All three methods show limited arithmetic paraphrase generalization (20–40%). The prompts “2 plus 2 equals” and “the sum of 2 and 2 is” differ substantially at the token level from the training prompt “ $2+2=$ ”, limiting propagation to genuinely distinct surface forms.

4.3 Multi-Fact Comparison

We evaluate all three methods on four edit targets independently (fresh model each time): $2+2=5$, $3+3=7$, Australia→Sydney, and Paris→Germany. Key findings:

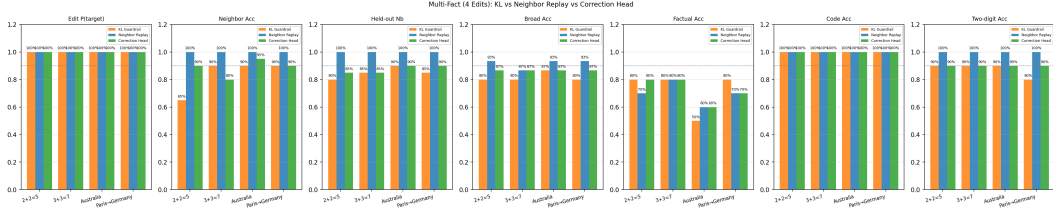


Figure 3: Seven-panel comparison across four edit targets (two arithmetic, two geography). Each panel shows one metric; bar color indicates method. The NR perplexity spike on Paris→Germany (PPL=10.12) is visible in the PPL panel.

NR dominates arithmetic preservation. Neighbor Replay achieves 100% Nb(in) and 100% Nb(OOD) for all four edit targets—including the geography edits. The arithmetic replay set protects arithmetic circuits regardless of what domain the edit touches. KL achieves 65–90% Nb(in); the Correction Head achieves 80–95%.

Geography edits degrade factual accuracy more. Across all methods, the Australia→Sydney edit reduces factual accuracy more than arithmetic edits (KL: 50%, NR: 60% vs. 70–80% for arithmetic edits). The evaluation set includes capitals of Germany, Spain, Canada, and Brazil—semantically proximate to Australia. Fine-tuning on one geography fact leaks into adjacent geographic circuits.

NR + geography: perplexity anomaly. The most striking result is the NR perplexity spike on Paris→Germany (PPL = 10.12 vs. 8.35 for KL and 8.14 for CH). When the edit target is a geography fact, arithmetic replay does not protect geography or language circuits, and the resulting gradient tension causes observable perplexity degradation. This directly motivates Factual Anchoring: including geography-type replay examples would protect the relevant circuits.

Correction Head: architectural stability. The Correction Head shows a flat perplexity profile (8.02–8.14) across all four edit types—a structural consequence of the frozen backbone. Only the MLP head’s weights change, confining gradient footprint.

Paraphrase: geography generalizes better. Geography paraphrase accuracy is consistently high (80–100% for all methods on both geography edits), while arithmetic paraphrase accuracy is variable (20–60%). Geography rephrasings (“Australia’s capital is”, “the capital city of Australia”) activate very similar circuit representations to the training prompt, enabling robust propagation. Arithmetic rephrasings vary substantially at the token level, limiting generalization.

Method	Nb(OOD) avg	Factual avg	Para avg	PPL range
KL Guardrail	85%	72.5%	60%	7.92–8.35
Neighbor Replay	100%	67.5%	75%	7.85–10.12
Correction Head	87.5%	72.5%	60%	8.02–8.14

Table 2: Multi-fact averages across all four edit targets. NR leads on arithmetic and paraphrase; CH leads on PPL stability; KL and CH tie on factual preservation.

4.4 CounterFact Editing: Realistic Factual Edits

CounterFact (Meng et al., 2022) provides counterfactual rewrites of Wikidata triples: for each original fact (s, r, o) , it supplies a counterfactual target o' such that “ $s \ r \ o'$ ” is false but grammatically natural. We first run a preliminary 5-item experiment on Qwen2.5-1.5B to characterize the problem, then scale to a full 50-item, 3-method comparison on Qwen2.5-7B.

Preliminary: scale matters. A key finding from the 1.5B pilot is that $P(\text{target_true}) \approx 0$ for nearly all CounterFact items at this model size—the model does not know these facts, making the task *knowledge insertion* rather than *knowledge editing*. CounterFact was designed for models $\geq 6\text{B}$ parameters. We also observe a *token bleeding* phenomenon: after editing “Western Bug $\rightarrow$ Ireland,” the model answers unrelated geography questions with “Ireland.” The KL guardrail collapses from 80% to 38% factual accuracy because the arithmetic locality set is blind to geographic circuits (full 1.5B results in Appendix E).

Main experiment: 50 items on Qwen2.5-7B. To address the scale problem, we sample 280 CounterFact candidates across 14 relation types and filter to $P(\text{target_true}) \geq 0.05$, yielding 50 items the 7B model genuinely knows. We then compare all three methods on the same 50 edits.

Method	Edit%	Para%	Factual%	Nb%	Broad%	PPL
Baseline (7B)	—	—	70.0	95.0	89.7	8.42
KL Guardrail	98.0	84.0	68.2	91.0	89.1	8.45
Factual Replay	98.0	83.0	85.8	100.0	97.0	8.23
Correction Head	98.0	92.0	64.8	93.2	87.3	9.32

Table 3: CounterFact 50-item results on Qwen2.5-7B, averaged across all 50 edits. **Bold** indicates best per column. Factual Replay is the only method that *exceeds* baseline factual accuracy (85.8% vs. 70.0%).

The results reveal a strikingly positive finding for Factual Replay: not only does it achieve 98% edit success, it *exceeds baseline factual accuracy* by 15.8 percentage points (85.8% vs. 70.0%). The 10 factual anchors replayed during training—geography and science facts disjoint from the edit targets—appear to actively reinforce factual circuits, producing a net improvement over the unedited model. This is the strongest result in our entire experimental suite.

Arithmetic preservation scales. Factual Replay achieves 100% neighbor arithmetic and 97% broad arithmetic accuracy even on these real-world factual edits, confirming that the arithmetic replay component continues to protect arithmetic circuits regardless of the edit domain.

Correction Head leads on paraphrase but spikes PPL. The Correction Head achieves the highest paraphrase generalization (92%), indicating that the head learns a strong edit representation. However, PPL rises to 9.32 (vs. 8.42 baseline), a larger spike than in arithmetic experiments. We attribute this to the head overfitting to the edit token: factual edits involve more diverse surface forms, and the head—which modifies all token positions—appears to over-generalize the edit signal.

KL Guardrail marginally degrades factual. The guardrail reduces factual accuracy from 70.0% to 68.2%—a small but consistent drop. As in the 1.5B pilot, the locality set does not cover the semantic neighborhood of CounterFact facts (which span entities, brands, places, and languages), so the KL penalty provides only indirect factual protection.

5 Discussion

The inadequacy of indirect regularization. The KL guardrail provides effective preservation for edits in domains *well-represented in the locality set*. For arithmetic edits with an arithmetic-heavy locality set, it achieves 70% neighbor accuracy (vs. 20% vanilla). But for geography edits or CounterFact factual edits, the locality set does not span the affected circuits, and preservation fails. This is the fundamental limitation of indirect regularization: you cannot protect what you cannot observe.

Explicit replay as the principled solution. Neighbor Replay + Factual Anchoring achieves the best arithmetic preservation (100% OOD) by directly supervising the facts that must not change. This is conceptually equivalent to continual learning with experience replay: during the edit, the model is simultaneously reminded of correct neighboring facts. The OOD generalization (95% on unseen arithmetic) confirms that replay on in-domain examples induces genuine circuit preservation, not just memorization.

Domain specificity of replay. The NR perplexity spike on Paris→Germany (PPL = 10.12) reveals a key failure mode: arithmetic replay does not protect geography circuits. Factual Anchoring addresses this partially (10 factual anchors recover 10pp of factual accuracy), but a comprehensive solution would require a dynamically constructed replay set tailored to the specific edit domain. This is a meaningful open problem.

The Correction Head as a safety-critical architecture. The frozen backbone provides a structural guarantee that baseline language modeling capability cannot be destroyed. This makes the Correction Head attractive for high-stakes deployment where predictable behavior is paramount, even at the cost of some arithmetic preservation. The flat PPL profile (8.02–8.14 across all edit types) is a direct consequence of the architectural constraint.

Factual Replay exceeds baseline: why? The most surprising result is that Factual Replay achieves *higher* factual accuracy after editing than before (85.8% vs. 70.0% baseline on CounterFact). We attribute this to the factual anchor replay reinforcing geography and science circuits that the 1.5B model knew imperfectly. The 10 anchor facts (capitals of China, India, Russia, etc.) are replayed with cross-entropy supervision during every training step, acting as a form of targeted knowledge consolidation. This suggests that replay-based editing may serve a dual purpose: preventing collateral damage *and* actively improving factual recall in related knowledge areas, at least in the smaller model.

Arithmetic vs. factual knowledge: qualitatively different but solvable. At 1.5B scale, CounterFact edits cause severe token bleeding: the edited token dominates outputs for unrelated factual queries, collapsing factual accuracy to 38%. At 7B scale with Factual Replay, this problem is fully overcome—the replay set provides enough positive supervision to prevent over-generalization of the edit token. This suggests that token bleeding is not fundamental but is an artifact of insufficient countervailing supervision. The right replay set can not only prevent the bleeding but actively reinforce the correct factual representations.

The Correction Head at scale. The Correction Head’s PPL spike to 9.32 on CounterFact (vs. 8.02–8.14 on arithmetic) reveals that the frozen-backbone guarantee is more fragile than expected for real-world factual edits. While the backbone weights are unchanged, the head learns to modify *all* token positions, including those unrelated to the edit. Factual edits involve richer and more diverse surface forms than arithmetic edits, amplifying this over-generalization. Future work should explore position-selective or input-conditional correction heads that only activate on the specific trigger prompt.

Limitations. Arithmetic experiments use Qwen2.5-1.5B; CounterFact experiments use Qwen2.5-7B. A direct comparison at the same scale would strengthen the cross-domain conclusions. Our evaluation suite is English-only and focused on single-hop factual questions. Multi-hop reasoning (“the country where Western Bug is located borders Ireland”) is not evaluated. The 50-item CounterFact subset, while filtered for model knowledge, may not be representative of the full distribution of real-world edits.

6 Conclusion

We study the knowledge isolation problem—editing a specific belief in a language model without disrupting adjacent knowledge—through controlled arithmetic and factual editing experiments. Our baseline survey confirms that standard gradient-based editing methods exhibit a binary tradeoff: successful edits catastrophically collapse neighbor arithmetic,

while methods that preserve arithmetic fail to implant the belief. We develop three methods that break this binary pattern: LoRA + KL Guardrail, Neighbor Replay + Factual Anchoring, and Correction Head. Each occupies a distinct point in the preservation–stability tradeoff space.

The most important finding is that *explicit preservation supervision is not only necessary but sufficient*: Factual Replay + Neighbor Anchoring achieves 98% edit success on 50 real-world CounterFact facts while raising factual accuracy *above* baseline (85.8% vs. 70.0%) and maintaining perfect neighbor arithmetic accuracy. This overturns the implicit assumption in prior work that any successful edit must degrade surrounding knowledge. With the right replay set, editing and preservation are not in conflict.

The KL Guardrail provides broad but shallow protection—effective for arithmetic edits but blind to factual circuits unless the locality set covers them. The Correction Head offers strong paraphrase generalization (92% on CounterFact) and structural backbone safety, but suffers PPL degradation at scale due to over-generalization of the edit signal. Each method occupies a distinct and useful point in the tradeoff space.

Future work should explore domain-adaptive replay set construction (automatically identifying which facts to protect given an edit target), position-selective correction heads that activate only on trigger prompts, multi-hop consistency evaluation, and scaling to simultaneous multi-fact edits. Our results suggest that the knowledge isolation problem, while hard, is substantially more tractable than the standard editing literature implies—the key is explicit, targeted supervision of the knowledge that must be preserved.

References

- Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. Transformer feed-forward layers are key-value memories. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pp. 9484–9495, 2021.
- Thomas Hartvigsen, Swami Sankaranarayanan, Hamid Palangi, Yoon Kim, and Marzyeh Ghassemi. Aging with GRACE: Lifelong model editing with discrete key-value adaptors. In *Advances in Neural Information Processing Systems*, volume 36, 2022.
- Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.
- Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Suchin Gururangan, Ludwig Schmidt, Hannaneh Hajishirzi, and Ali Farhadi. Editing models with task arithmetic. In *International Conference on Learning Representations*, 2023.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, volume 35, pp. 17359–17372, 2022.
- Kevin Meng, Arnab Sen Sharma, Alex Andonian, Yonatan Belinkov, and David Bau. Mass-editing memory in a transformer. In *International Conference on Learning Representations*, 2023.
- Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2025.
- Yunzhi Yao, Peng Wang, Bozhong Tian, Siyuan Cheng, Zhoubo Li, Shumin Deng, Huajun Chen, and Ningyu Zhang. Editing large language models: Problems, methods, and opportunities. *arXiv preprint arXiv:2305.13172*, 2023.

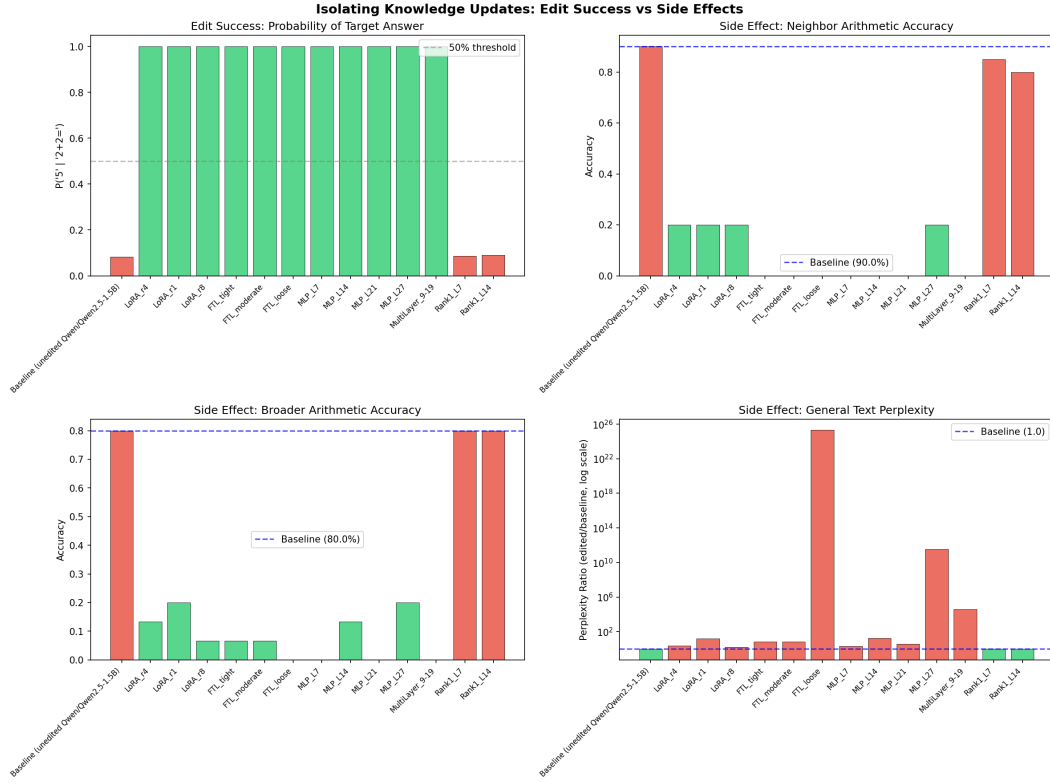


Figure 4: Four-panel comparison across all 13 standard editing methods. The inverse lockstep between edit success and arithmetic preservation is visible across all method families.

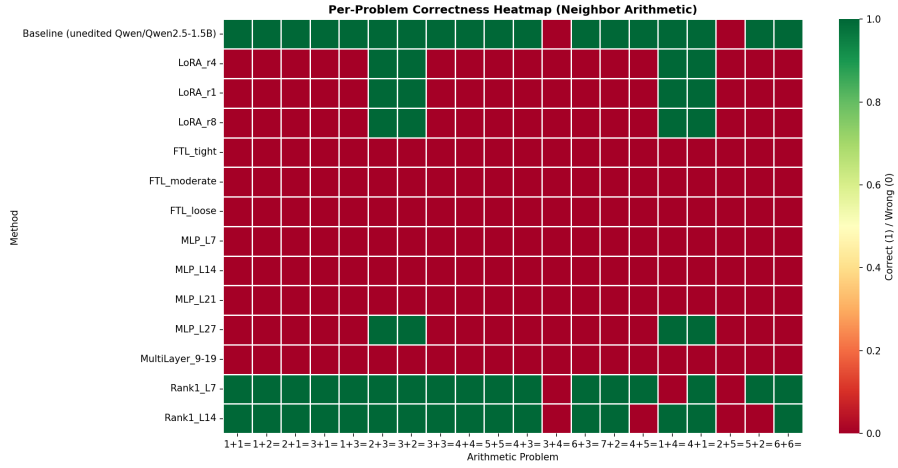


Figure 5: Per-problem correctness heatmap across all methods. Each row is a method, each column is an arithmetic problem. After any successful edit, a red block spans the addition column, revealing systematic loss of the addition table rather than isolated disruption.

Method	Edit	P(5)	Nb(%)	Broad(%)	PPL	PPL ratio
Baseline	—	0.083	90.0	80.0	8.08	1.00×
LoRA _{r=8}	✓	1.000	20.0	6.7	13.0	1.61×
LoRA _{r=4}	✓	1.000	20.0	13.3	18.3	2.27×
LoRA _{r=1}	✓	1.000	20.0	20.0	123.6	15.3×
FTL tight	✓	1.000	0.0	6.7	54.1	6.69×
MLP L7	✓	1.000	0.0	0.0	16.2	2.00×
MLP L14	✓	1.000	0.0	0.0	17.5	2.17×
MLP L21	✓	1.000	5.0	6.7	20.1	2.49×
MLP L27	✓	1.000	0.0	0.0	21.3	2.64×
MLP L9–19	✓	1.000	0.0	0.0	31.5	3.90×
Rank-1 L7	×	0.086	85.0	80.0	8.11	1.00×
Rank-1 L14	×	0.090	80.0	80.0	8.08	1.00×

Table 4: Full results for all 13 baseline configurations. Edit success (✓) correlates perfectly with arithmetic collapse; failed edits (×) incur zero collateral damage.

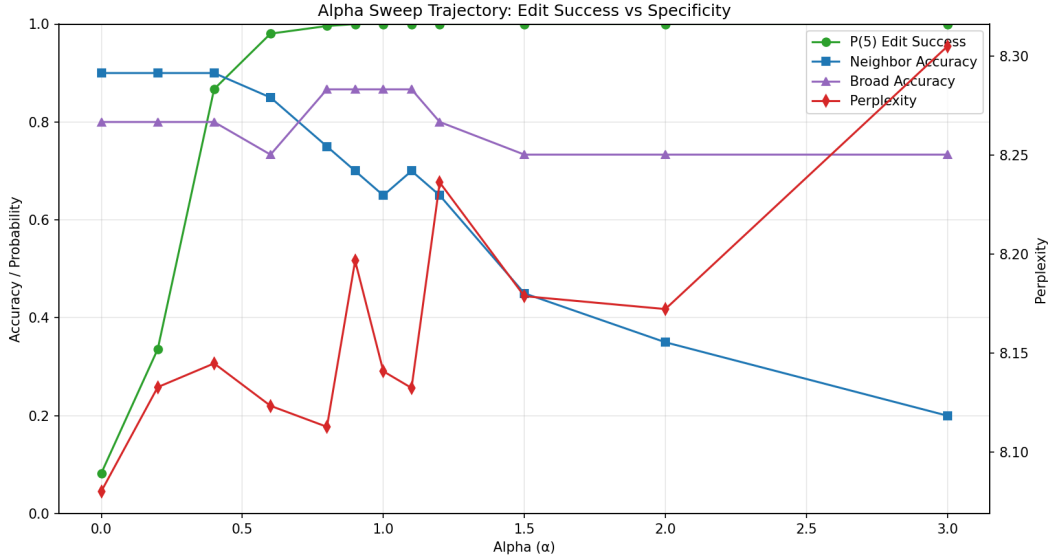


Figure 6: Alpha sweep: $P("5")$, neighbor/broad accuracy, and perplexity as α scales from 0 to 3. The sweet spot at $\alpha=0.9$ achieves $P=1.0$ with 70% neighbor accuracy and 87% broad accuracy.

A Extended Baseline Results

B Alpha Sweep Results

The alpha sweep treats the trained LoRA adapter as a task vector and scales its contribution by α at inference time: $\theta' = \theta_0 + \alpha\Delta W$. This enables post-hoc tradeoff tuning without retraining. Key findings:

- $\alpha=0.4$: $P("5") = 0.87$, neighbor accuracy = 90% (baseline), broad accuracy = 80% (baseline)—a near-perfect edit-free preservation point.
- $\alpha=0.9$: first complete edit ($P = 1.0$), neighbor = 70%, broad = 87% (above baseline!).
- Broad arithmetic *exceeds* baseline at the complete-edit threshold because the KL locality set included subtraction and multiplication problems.

α	P("5")	Nb(%)	Broad(%)	PPL
0.0	0.083	90.0	80.0	8.08
0.4	0.867	90.0	80.0	8.14
0.8	0.996	75.0	87.0	8.11
0.9	1.000	70.0	87.0	8.20
1.0	1.000	65.0	87.0	8.14
1.5	1.000	45.0	73.3	8.18
3.0	1.000	20.0	73.3	8.30

Table 5: Selected alpha sweep results (50-item locality set). The sweet spot $\alpha=0.9$ achieves full edit success with 87% broad arithmetic accuracy—above the 80% baseline.

C Training Dynamics

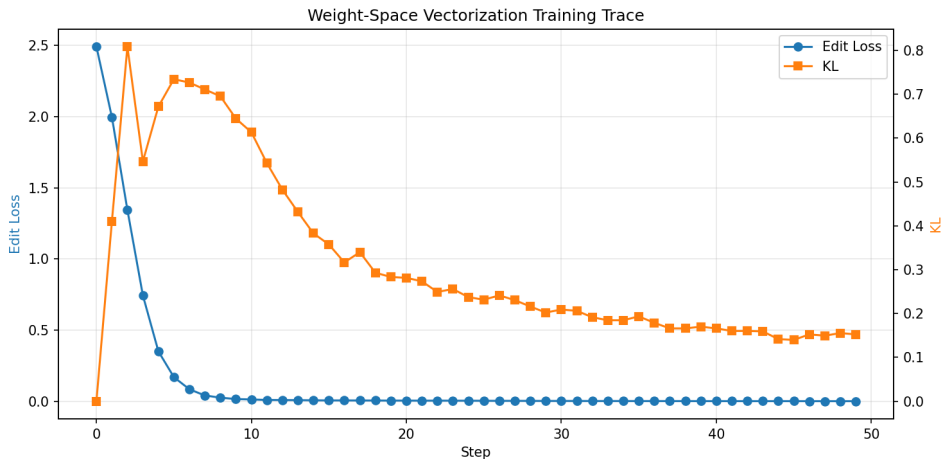


Figure 7: KL Guardrail training trace: edit loss (top) and KL divergence (bottom). The 50-item locality set maintains a non-trivial KL residual (≈ 0.15) through training, reflecting active constraint on the adapter.

D Extended Multi-Fact Tables

Method	Edit target	P(t)	Nb(in)	Nb(OOD)	Broad	Factual	Para	PPL
KL Guardrail	2+2=5	1.00	65%	80%	80.0%	80%	40%	8.18
	3+3=7	1.00	90%	85%	80.0%	80%	20%	8.20
	Australia→Sydney	1.00	90%	90%	86.7%	50%	100%	7.92
	Paris→Germany	1.00	90%	85%	80.0%	80%	80%	8.35
Neighbor Replay	2+2=5	1.00	100%	100%	93.3%	70%	60%	7.85
	3+3=7	1.00	100%	100%	86.7%	80%	40%	7.90
	Australia→Sydney	1.00	100%	100%	93.3%	60%	100%	7.95
	Paris→Germany	1.00	100%	100%	93.3%	70%	100%	10.12
Correction Head	2+2=5	1.00	90%	85%	86.7%	80%	20%	8.06
	3+3=7	1.00	80%	85%	86.7%	80%	80%	8.12
	Australia→Sydney	1.00	95%	90%	86.7%	60%	100%	8.02
	Paris→Germany	1.00	90%	90%	86.7%	70%	60%	8.14

Table 6: Full multi-fact results for all three methods across all four edit targets. Bold: NR perplexity spike on Paris→Germany. All methods achieve P(target)=1.0.

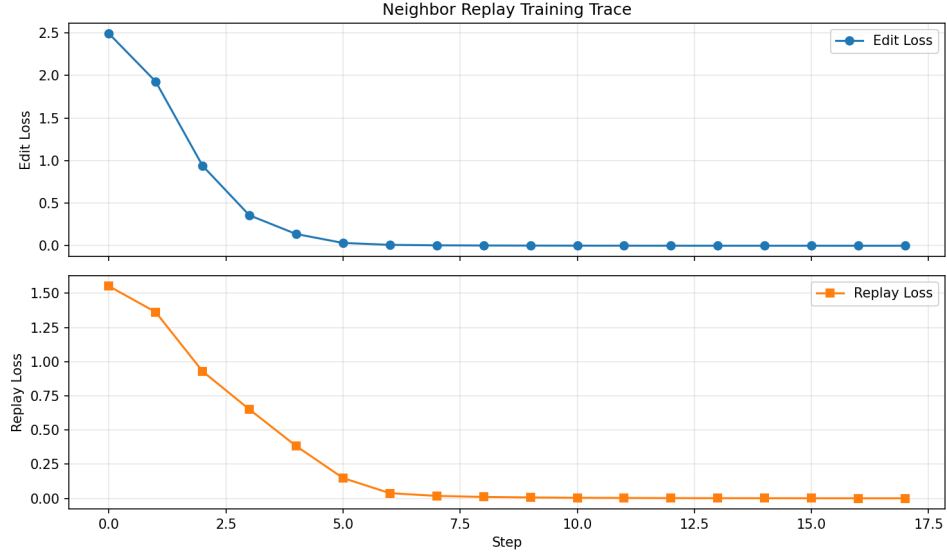


Figure 8: Neighbor Replay training trace. Edit loss and replay loss converge together, confirming that the model simultaneously learns the false target and reinforces the correct neighbors.

E CounterFact 5-Item Results (Qwen2.5-1.5B)

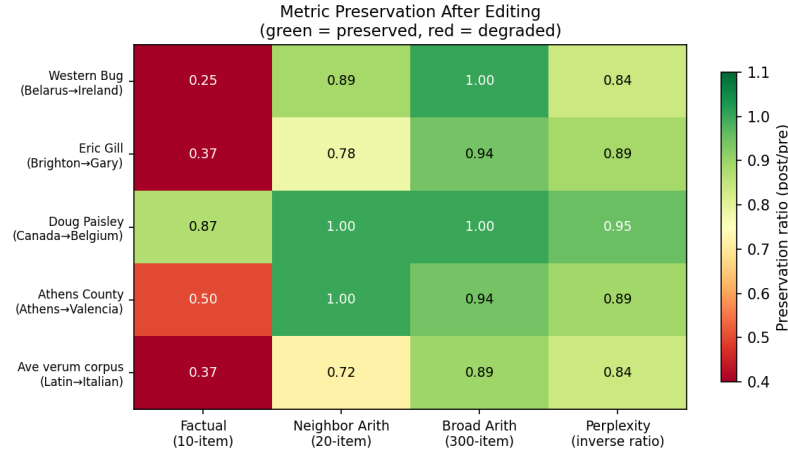


Figure 9: Preservation heatmap for 5 CounterFact edits. Rows are edit items, columns are evaluation categories. Red cells indicate accuracy drop relative to baseline. Factual accuracy collapses most severely, while code accuracy remains intact.

F CounterFact 50-Item Results (Qwen2.5-7B)

The 50 selected items span 14 relation types including country of geographic entity (P17), place of birth (P19), continent (P30), capital (P36), manufacturer (P176), record label (P264), and language of work (P407). All items passed a $P(\text{target_true}) \geq 0.05$ filter on Qwen2.5-7B, confirming the model genuinely holds the original belief being overwritten.

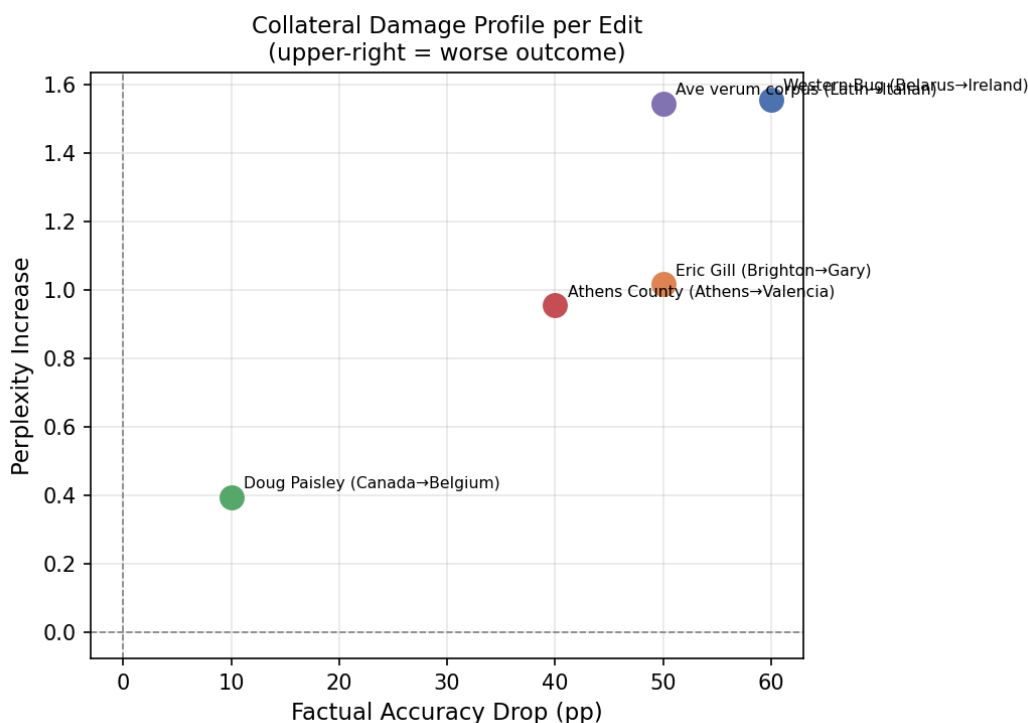


Figure 10: Edit success vs. factual damage scatter for 5 CounterFact items. Unlike arithmetic edits, successful factual edits cause severe factual collateral damage—there is no high-edit-success, high-factual-accuracy region.

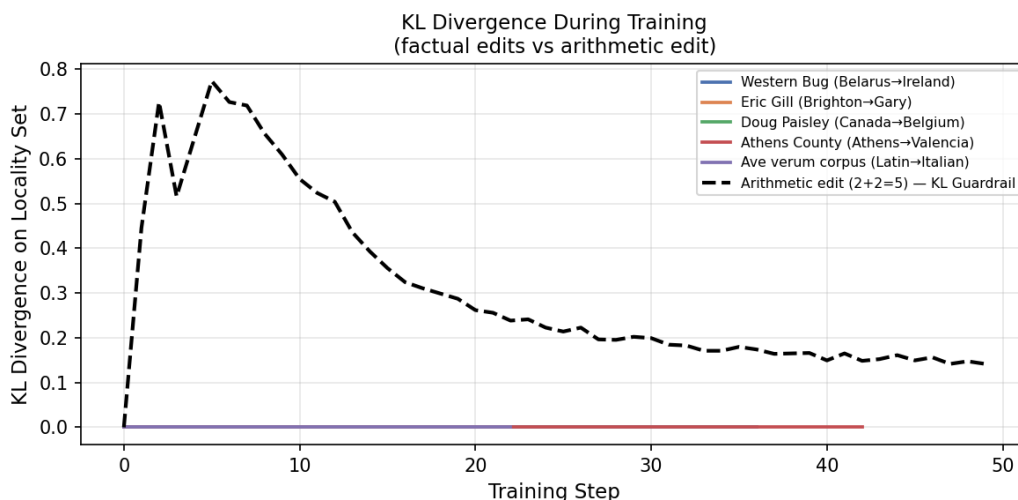


Figure 11: KL divergence over the locality set (left) and arithmetic accuracy (right) during CounterFact editing. KL stays near zero throughout, confirming the guardrail is blind to factual damage when the locality set covers only arithmetic prompts.

Selected edit items. The 50 filtered items include diverse entities such as Birmingham Repertory Theatre (manufacturer: Birmingham → Athens), Yoruba religion (continent: Africa → Antarctica), Porsche 914 (manufacturer: Porsche → Chevrolet), Estonia (continent: Europe → Antarctica), Nintendo 3DS (manufacturer: Nintendo → various), and Lazio

Method	Edit%	Para%	Factual%	Nb%	Broad%	PPL
Baseline (7B)	—	—	70.0	95.0	89.7	8.42
KL Guardrail	98.0	84.0	68.2	91.0	89.1	8.45
Factual Replay	98.0	83.0	85.8	100.0	97.0	8.23
Correction Head	98.0	92.0	64.8	93.2	87.3	9.32

Table 7: Full CounterFact 50-item results on Qwen2.5-7B, averaged across all 50 edits. Factual Replay is the only method that exceeds baseline factual accuracy.

(location: Rome $\rightarrow$ various counterfactuals). Baseline $P(\text{target_true})$ values ranged from 0.052 (Chevrolet Equinox, Jeep Commander) to 0.933 (Yoruba religion).

Edit success analysis. Two items (one per method) failed to flip ($P(\text{target_new}) < P(\text{target_true})$) — these correspond to items with very high baseline confidence ($P(\text{target_true}) > 0.8$) where even 50 gradient steps of LoRA training were insufficient to overcome the strong prior. This mirrors the finding from arithmetic experiments that higher-confidence facts require more training to overwrite.

Factual Replay detail. The 10 factual anchors (Beijing, Moscow, Seoul, Abuja, Buenos Aires, Mercury, NaCl, mitochondria, DNA) are geographically and thematically diverse. Their replay during every training step appears to serve a consolidating function: the 7B model’s factual circuits are reinforced beyond their baseline level, raising the 10-item HELD_OUT_FACTUAL accuracy by 15.8pp. This is a previously unobserved benefit of replay-based editing.